

Adaptive Local-Global Similarity for Comparing Heterogeneous Unordered Embedding Sets

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Abstract

Many information-system and machine-learning tasks compare finite unordered sets of embeddings rather than individual vectors. Existing similarity measures often emphasize either local neighborhood structure or global distributional structure, although the two properties are complementary. We define Adaptive Local-Global similarity (ALG), a symmetric similarity measure that combines a local similarity with a global similarity. The local and global similarities and the resulting configurable ALG similarity take values in $[0, 1]$; the additive Initial ALG reference instance is not constrained to this interval. The local similarity uses adaptive k -nearest-neighbor scales, while the global similarity is obtained by normalizing a distributional discrepancy. We evaluate ALG on 243 benchmark datasets from seven dataset families (graphs, images, OpenML, tabular data, text, time series, and neural-model activations). The 243-dataset benchmark is divided into 60 training datasets and 186 held-out test datasets. A single configuration is selected on the 60 training datasets and then fixed before evaluation on the 186 held-out test datasets. The selected configuration reaches a mean positive-label F1-score across datasets of 0.7556, compared with 0.7498 for Energy distance, the strongest literature baseline with complete test coverage. Ablation and controlled perturbation experiments further characterize the complementarity of the local and global similarities and their response to controlled distributional changes. These results support local-global similarity as a reliable approach for comparing heterogeneous embedding sets.

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1. Introduction

Many data-integration and machine-learning pipelines represent an object by a finite set of vectors. A database attribute can be represented by embeddings of its name and values, a graph by node embeddings, an image by local descriptors, a text by token or sentence embeddings, and a time series by embeddings of sliding windows. Neural-network layers can also be represented by sets of hidden activations. In these settings, the comparison problem is not the comparison of two individual vectors, but the comparison of two finite unordered embedding sets.

A similarity measure between embedding sets must account for more than one property of the representation. Neighborhood-based measures describe local support and density, but they can remain high when two embedding sets undergo a coherent displacement. Global measures describe aggregate or distributional differences, but a global summary can assign similar values to embedding sets with different local structure. Kernel two-sample measures such as MMD compare empirical distributions through a reproducing-kernel Hilbert space [1]; optimal-transport measures compare the mass displacement required to transform one empirical distribution into another [2]; and nearest-neighbor measures estimate local support from within-set neighborhoods [3, 4, 5]. These measures capture different properties of the same embedding sets.

Figure 1 illustrates the resulting limitation. Two embedding sets can have similar global summaries while having different local support. In the opposite case, two embedding sets can preserve similar local neighborhoods while being globally displaced. A similarity measure based only on one of these properties can therefore miss a difference that is explicit in the other.

The same issue is central to the evaluation of generative models and point-set representations. Kynkäänniemi et al. [3] use sample-dependent nearest-neighbor radii to estimate local support. Naeem et al. [4] define Density and Coverage after identifying failure cases of earlier precision-recall estimators. Cheema and Uner [5] formalize bidirectional nearest-neighbor coverage, while Khayatkhoei and AbdAlmageed [6] show that nearest-neighbor precision and recall can become asymmetric in high dimension. For point sets, Chamfer

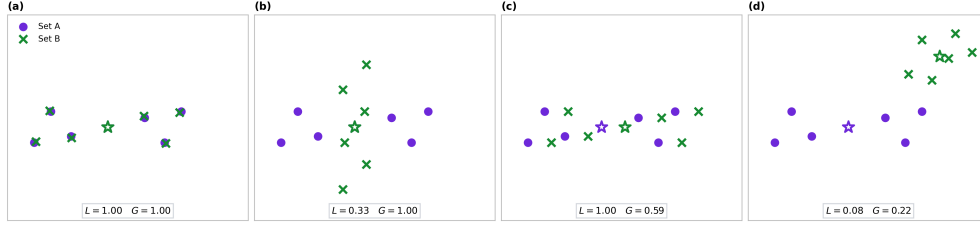


Figure 1: Local and global similarity capture complementary properties of two embedding sets. A global summary can remain similar when local support changes, whereas a local neighborhood similarity can remain high after a coherent global displacement. ALG combines the two similarities explicitly.

distance, Earth Mover’s Distance, and Density-Aware Chamfer Distance retain different geometric information [7, 8]. These results motivate a similarity measure in which local neighborhood structure and global distributional difference are both explicit.

We first instantiate this idea as *Initial ALG*: symmetric cosine neighborhood coverage with $k = 1$ plus cosine similarity between pooled centroids. Initial ALG exposes the local-global principle but fixes the local similarity, global similarity, geometry, and combination rule. We therefore define *Adaptive Local-Global similarity* (ALG) as the general configurable formulation. The configuration selected only on the training datasets and then frozen is denoted ALG*.

The empirical evaluation separates configuration selection from final evaluation. Configuration selection is performed on 60 training datasets, one ALG configuration is selected, and that configuration is then fixed for evaluation on 186 held-out test datasets. The evaluation covers 243 benchmark datasets and a search space of 168,960 ALG configurations; the frozen configuration is also evaluated on three additional neural-model-activation datasets. The evaluation compares ALG with 38 literature baselines and two standard pooled baselines, while internal ALG components are reported separately in ablation analyses.

The contributions of this work are the following:

- We define Initial ALG as the reference local-global instance and generalize it into ALG, a symmetric and permutation-invariant formulation for finite unordered embedding sets. ALG combines a candidate local similarity with a normalized candidate global similarity in a common range.

- We establish a unified configuration-selection and evaluation protocol across seven heterogeneous dataset families. The held-out test evaluation spans seven of these families. Our benchmark brings together 38 literature baselines from different research domains, including methods specifically designed for particular data families, alongside general-purpose similarity measures. ALG and all tunable baselines are configured exclusively on the same 60 training datasets, with their selected configurations fixed before evaluation on the same 186 held-out test datasets in the completed evaluation. This protocol enables a consistent comparison of domain-specific and general-purpose methods across the evaluated dataset families, without dataset-specific or test-time configuration selection.
- We provide a large-scale empirical analysis of local-global similarity across heterogeneous embedding sets. The evaluation includes literature and pooled baselines, matched local-only and global-only ablations, cross-family analysis, controlled perturbation experiments, and computational-efficiency measurements. These experiments characterize when local and global similarities are complementary and how the compared similarity measures respond to controlled distributional changes.

The remainder of the paper is organized as follows. Section 2 reviews neighborhood-based, distributional, and point-set similarity measures. Section 3 defines ALG. Section 4 describes the experimental methodology. Section 5 reports and discusses the results, and Section 6 concludes the paper.

2. Related work

2.1. Neighborhood-based similarity

Nearest-neighbor methods estimate local properties directly from empirical samples. Kynkäänniemi et al. [3] represent an empirical distribution by the union of hyperspheres centered on the observed samples. The radius of each hypersphere is determined by a within-sample k -nearest-neighbor distance, so the neighborhood scale changes with local density. Precision and recall are then obtained by testing whether samples from one distribution fall inside the empirical support of the other.

Naeem et al. [4] analyze failure cases of these estimators and define Density and Coverage to improve their reliability under outliers and changes

in neighborhood structure. Cheema and Uner [5] define Precision Recall Cover using bidirectional nearest-neighbor relations and study its mathematical properties. Khayatkhoei and AbdAlmageed [6] show that nearest-neighbor precision and recall can exhibit systematic asymmetry as dimension increases. These works establish both the usefulness and the limitations of adaptive local support. ALG uses the same general principle for its local similarity: the scale of a cross-set comparison is defined from within-set neighborhoods and the local similarity is symmetrized.

2.2. Global distributional similarity

Global two-sample measures compare empirical distributions without reducing the comparison to nearest-neighbor coverage. Maximum Mean Discrepancy (MMD) compares distributions through differences in expectations over a reproducing-kernel Hilbert space [1, 9]. Its empirical behavior depends on the kernel and its bandwidth. Energy distance provides a distance between distributions through expected pairwise distances, while Fréchet-type measures summarize empirical distributions by low-order moments. These measures capture global distributional differences but do not preserve the same local neighborhood information as nearest-neighbor measures.

Optimal transport compares two empirical distributions through the cost of transporting mass between them. Exact transport is computationally expensive; entropic regularization and the Sinkhorn algorithm provide a practical approximation [2], and sliced Wasserstein measures reduce the computational burden through one-dimensional projections. In ALG, MMD, Energy distance, Fréchet distance, centroid distance, and sliced Wasserstein distance are treated as candidate global discrepancies that are normalized to a common similarity range before being combined with the local similarity.

2.3. Point-set distances

Point-set distances provide another family of comparisons between finite sets. Chamfer distance aggregates nearest-neighbor distances in both directions, but many-to-one assignments can make it insensitive to density differences. Earth Mover’s Distance uses a global assignment and therefore retains different information from Chamfer distance. Density-Aware Chamfer Distance (DCD) adds a density correction to the nearest-neighbor comparison and bounds the resulting measure [8]. The ALG configuration space includes a DCD-inspired local similarity together with coverage, manifold precision-recall F1, best buddies, and local scaling.

130 2.4. *Embedding-set comparison*

131 Data integration systems often rely on multiple similarity signals to handle
 132 heterogeneous schemas and data representations. Classical schema matching
 133 systems such as Cupid, COMA, and Similarity Flooding combine lexical,
 134 instance-level, and structural evidence [10, 11, 12]. More recent embedding-
 135 based approaches improve the semantic representation of schema elements and
 136 data values [13, 14, 15, 16]. Across these systems, combining complementary
 137 sources of information is typically more effective than relying on a single
 138 signal, because different signals capture different aspects of heterogeneity.

139 This observation also applies once objects are represented as sets of em-
 140 beddings. A remaining question is how two such sets should be compared
 141 without collapsing their structure into a single representation or relying on
 142 only one geometric criterion. ALG follows the same principle of comple-
 143 mentary evidence by combining local neighborhood information with global
 144 distributional information in a single similarity measure.

145 3. Adaptive Local-Global similarity

146 3.1. *Problem formulation*

147 Let

$$A = \{a_i\}_{i=1}^n, \quad B = \{b_j\}_{j=1}^m, \quad a_i, b_j \in \mathbb{R}^D,$$

148 be two finite unordered embedding sets, with $n, m \geq 2$.

149 Our objective is to compare A and B using two complementary sources of
 150 information: their local neighborhood structure and their global distributional
 151 structure. Since the embeddings are unordered, the similarity must be
 152 permutation invariant. Moreover, because neither set is treated as a reference,
 153 the similarity must be symmetric.

154 We first introduce the initial formulation that motivated ALG. We then
 155 generalize this formulation by allowing the local similarity, the global similarity,
 156 their underlying geometries, and their relative contributions to vary.

157 3.2. *Initial ALG*

158 Initial ALG combines a local similarity based on bidirectional neighbor-
 159 hood coverage with a global similarity based on the cosine similarity between
 160 the two set centroids.

161 *Local similarity.* Let d be a non-negative dissimilarity. For each $a_i \in A$, we
 162 define its within-set k -nearest-neighbor radius as

$$\rho_A^{(k)}(a_i) = d(a_i, \text{kNN}_k(a_i; A \setminus \{a_i\})), \quad (1)$$

163 and define $\rho_B^{(k)}$ analogously.

164 The directed coverage from A to B is the proportion of points in A whose
 165 nearest point in B lies within their within-set neighborhood radius:

$$\text{Cov}_d^{(k)}(A \rightarrow B) = \frac{1}{n} \sum_{i=1}^n \mathbf{1} \left[\min_{b \in B} d(a_i, b) \leq \rho_A^{(k)}(a_i) \right]. \quad (2)$$

166 The same quantity is computed from B to A , and the two directions are
 167 averaged to obtain the symmetric coverage similarity:

$$L_{\text{cov},d}^{(k)}(A, B) = \frac{1}{2} \left[\text{Cov}_d^{(k)}(A \rightarrow B) + \text{Cov}_d^{(k)}(B \rightarrow A) \right]. \quad (3)$$

168 The neighborhood radius is estimated from each embedding set itself
 169 and therefore adapts to local density. This principle is related to empirical
 170 precision, recall, density, and coverage measures [3, 4, 5].

171 Initial ALG uses $k = 1$ and cosine dissimilarity. For nonzero vectors x
 172 and y , cosine dissimilarity is defined as

$$\delta_{\text{cos}}(x, y) = 1 - \frac{x^\top y}{\|x\|_2 \|y\|_2}. \quad (4)$$

173 Thus, the local similarity measures the proportion of points from each set
 174 that are covered by the other set relative to the local neighborhood scale of
 175 the original set.

176 *Global similarity.* The global similarity first summarizes each embedding set
 177 by its centroid:

$$\mu_A = \frac{1}{n} \sum_{i=1}^n a_i, \quad \mu_B = \frac{1}{m} \sum_{j=1}^m b_j. \quad (5)$$

178 The global similarity is then defined as the cosine similarity between these
 179 two centroids:

$$S_{\text{cos}}^{\text{cent}}(A, B) = \frac{\mu_A^\top \mu_B}{\|\mu_A\|_2 \|\mu_B\|_2}. \quad (6)$$

180 Initial ALG combines the local and global similarities as

$$\text{ALG}_{\text{init}}(A, B) = L_{\text{cov}, \delta_{\text{cos}}}^{(1)}(A, B) + S_{\text{cos}}^{\text{cent}}(A, B). \quad (7)$$

181 This initial formulation captures the central intuition of ALG: two em-
 182 bedding sets should be compared using both local and global information.
 183 However, it fixes one particular local similarity, one particular global similar-
 184 ity, and a fixed combination of the two. Moreover, because centroid cosine
 185 similarity takes values in $[-1, 1]$, Initial ALG is not constrained to $[0, 1]$.

186 We therefore generalize this formulation into a bounded and configurable
 187 local-global similarity.

188 3.3. ALG formulation

189 Let

$$0 \leq L(A, B) \leq 1 \quad \text{and} \quad 0 \leq G(A, B) \leq 1$$

190 denote a bounded local similarity and a bounded global similarity, respectively.

191 ALG is defined as

$$\text{ALG}_{L,G,\lambda}(A, B) = \lambda L(A, B) + (1 - \lambda)G(A, B), \quad \lambda \in [0, 1]. \quad (8)$$

192 We refer to λ as the *local-global weight*. It controls the relative contribution
 193 of the two similarities. The endpoints $\lambda = 1$ and $\lambda = 0$ correspond respectively
 194 to local-only and global-only similarities, whereas intermediate values combine
 195 both sources of information.

196 **Proposition 1** (Structural properties). *Suppose that L and G are symmetric,*
 197 *permutation invariant, and bounded in $[0, 1]$. Then, for every $\lambda \in [0, 1]$,*
 198 *$\text{ALG}_{L,G,\lambda}$ is symmetric, permutation invariant, and bounded in $[0, 1]$.*

199 *Proof.* Symmetry and permutation invariance are preserved by scalar mul-
 200 tiplication and addition. Boundedness follows because Eq. (8) is a convex
 201 combination of two values in $[0, 1]$. $\square$

202 Rather than fixing L and G a priori, we evaluate several candidate local
 203 and global similarities and select their configuration using only the training
 204 datasets.

205 *3.4. Candidate local similarities*

206 The local similarity captures how similarly the two embedding sets are
207 organized at the neighborhood level.

208 For neighborhood-based similarities, the radius defined in Eq. (1) provides
209 an adaptive local scale. It is smaller in dense regions and larger in sparse
210 regions, allowing distances between the two sets to be interpreted relative to
211 their local neighborhood structure.

212 Coverage, defined in Eq. (3), is the local similarity used in Initial ALG.
213 ALG also considers alternative local similarities.

214 One such candidate is the *local-scaling similarity*. Let

$$j(i) = \arg \min_j d(a_i, b_j)$$

215 denote the nearest point in B to a_i , and define $i(j)$ analogously for the reverse
216 direction. The directed local-scaling similarity from A to B is

$$L_{A \rightarrow B, d}^{(k)} = \frac{1}{n} \sum_{i=1}^n \exp \left(- \frac{d(a_i, b_{j(i)})^2}{\rho_A^{(k)}(a_i) \rho_B^{(k)}(b_{j(i)})} \right). \quad (9)$$

217 The symmetric local-scaling similarity is

$$L_{\text{LS}, d}^{(k)}(A, B) = \frac{1}{2} \left(L_{A \rightarrow B, d}^{(k)} + L_{B \rightarrow A, d}^{(k)} \right). \quad (10)$$

218 The denominator normalizes the distance between matched points using
219 their within-set neighborhood scales. Consequently, the same absolute dis-
220 placement receives a larger penalty in a dense region than in a sparse region.
221 This construction is related to self-tuning neighborhood kernels [17].

222 The *Manifold PR-F1* candidate uses the same within-set radii to form
223 two empirical manifolds. Its precision and recall are

$$P_d^{(k)}(A, B) = \frac{1}{m} \sum_{j=1}^m \mathbf{1} \left[\exists i : d(a_i, b_j) \leq \rho_A^{(k)}(a_i) \right], \quad (11)$$

$$R_d^{(k)}(A, B) = \frac{1}{n} \sum_{i=1}^n \mathbf{1} \left[\exists j : d(a_i, b_j) \leq \rho_B^{(k)}(b_j) \right]. \quad (12)$$

224 The local similarity is their harmonic mean, $L_{\text{MPR}, d}^{(k)} = 2PR/(P + R)$, with
225 value zero when $P + R = 0$.

226 For *Best Buddies*, let $j(i)$ be the nearest point in B to a_i and $i(j)$ the
 227 nearest point in A to b_j . The similarity is the fraction of mutual nearest-
 228 neighbor pairs normalized by the smaller set:

$$L_{\text{BB},d}(A, B) = \frac{1}{\min(n, m)} \sum_{i=1}^n \mathbf{1}[i(j(i)) = i]. \quad (13)$$

229 The *DCD-inspired* candidate additionally penalizes many-to-one nearest
 230 neighbor assignments. Define $c_B(j) = \max\{1, \sum_i \mathbf{1}[j(i) = j]\}$ and $c_A(i) =$
 231 $\max\{1, \sum_j \mathbf{1}[i(j) = i]\}$. For $\alpha > 0$,

$$L_{\text{DCD},d}^{(\alpha)}(A, B) = \frac{1}{2} \left[\frac{1}{n} \sum_i \frac{e^{-\alpha d(a_i, b_{j(i)})^2}}{c_B(j(i))} + \frac{1}{m} \sum_j \frac{e^{-\alpha d(a_{i(j)}, b_j)^2}}{c_A(i(j))} \right]. \quad (14)$$

232 The complete search space contains five local-similarity families and four
 233 base dissimilarities: Euclidean, Manhattan, Chebyshev, and cosine dissimilar-
 234 ity.

Table 1: Local-similarity search space.

Local similarity	Parameters	Count
Coverage	4 dissimilarities $\times k \in \{1, 3, 5, 10\}$	16
Local scaling	4 dissimilarities $\times k \in \{1, 3, 5, 10\}$	16
Manifold PR-F1	4 dissimilarities $\times k \in \{1, 3, 5, 10\}$	16
DCD-inspired	4 dissimilarities $\times \alpha \in \{10, 50, 100\}$	12
Best Buddies	4 dissimilarities	4
Total		64

235 The neighborhood parameter k is included whenever the corresponding
 236 local similarity depends on a neighborhood radius.

237 3.5. Candidate global similarities

238 The global similarity is obtained from a *global discrepancy*. We define
 239 a global discrepancy $\Delta_g(A, B) \geq 0$ as a non-negative quantity measuring
 240 the overall difference between the two embedding sets according to a chosen
 241 distributional or geometric criterion. Because a global discrepancy increases
 242 as the two sets become less similar, it must be transformed before being
 243 combined with the local similarity.

244 We consider eight global discrepancies. Four compare the centroids μ_A
 245 and μ_B using Euclidean, Manhattan, Chebyshev, or cosine dissimilarity. The
 246 remaining global discrepancies are Energy distance, RBF-MMD, Gaussian
 247 Fréchet distance, and sliced Wasserstein distance.

Table 2: Global-discrepancy search space before normalization.

Global discrepancy	Geometry or specification	Count
Centroid	Euclidean, Manhattan, Chebyshev, cosine dissimilarity	4
Energy	Euclidean pairwise geometry	1
RBF-MMD	RBF kernel with median-bandwidth rule	1
Gaussian Fréchet	Mean and covariance discrepancy	1
Sliced Wasserstein	Euclidean projected transport	1
Total		8

248 The global discrepancies have different numerical ranges and therefore can-
 249 not be directly combined with the local similarities. Each global discrepancy
 250 $\Delta_g(A, B)$ is transformed into a global similarity:

$$G_{g,\psi,s}(A, B) = \psi\left(\frac{\Delta_g(A, B)}{s \sigma_g(A, B)}\right), \quad (15)$$

251 where $\sigma_g(A, B) > 0$ is a reference scale, $s > 0$ is a scale multiplier, and

$$\psi : [0, \infty) \rightarrow [0, 1]$$

252 is a monotonically decreasing normalization function.

253 We evaluate six normalization families. The clipped linear normalization
 254 is geometry-specific:

$$\psi_{\text{lin},\cos}(z) = \max(0, 1 - z/2), \quad (16)$$

$$\psi_{\text{lin},d}(z) = \max(0, 1 - z), \quad d \in \{L_2, L_1, L_\infty\}. \quad (17)$$

255 The remaining five normalization functions are

$$\frac{1}{1+z}, \quad \frac{1}{1+z^2}, \quad e^{-z}, \quad e^{-z^2/2}, \quad \frac{2}{1+e^z},$$

256 corresponding respectively to rational, Cauchy, exponential, Gaussian, and
 257 logistic normalization.

258 Each normalization is evaluated with

$$s \in \{0.25, 0.5, 1, 2, 4\}.$$

259 Consequently, each global discrepancy produces

$$6 \times 5 = 30$$

260 normalized global similarities. With eight global discrepancies, this gives

$$8 \times 30 = 240$$

261 candidate global similarities.

262 For centroid-based global discrepancies,

$$\Delta_{\text{cent},d}(A, B) = d(\mu_A, \mu_B). \quad (18)$$

263 For cosine centroid discrepancy,

$$\Delta_{\text{cent},\cos}(A, B) = \delta_{\cos}(\mu_A, \mu_B),$$

264 with $\sigma_g = 1$.

265 Let $r(A, B)$ denote the median of the pooled within-set one-nearest-
 266 neighbor distances in A and B , lower-bounded by machine precision. The
 267 reference scale is $\sigma_g = r(A, B)$ for centroid Euclidean, Manhattan, and
 268 Chebyshev discrepancies, Energy distance, and sliced Wasserstein distance;
 269 $\sigma_g = 1$ for centroid cosine discrepancy and RBF-MMD; and $\sigma_g = r(A, B)^2$
 270 for Gaussian Fréchet distance. Sliced Wasserstein uses 32 unit-normalized
 271 Gaussian projection directions generated with random seed zero. For every
 272 neighborhood-based local similarity, the effective value is $\min(k, n-1)$ within a
 273 set of size n . Cosine distances involving a zero vector are assigned the neutral
 274 dissimilarity value one, and denominators are lower-bounded by machine
 275 precision.

276 3.6. Configuration space and selection

277 ALG jointly varies the local similarity, the global similarity, and the
 278 local-global weight.

279 The search space contains

- 280 • 64 local similarities;

- 281 • 240 global similarities;
 - 282 • 11 local-global weights, $\lambda \in \{0, 0.1, \dots, 1\}$.
- 283 The complete search space therefore contains

$$64 \times 240 \times 11 = 168,960 \quad (19)$$

284 ALG configurations.

285 All configuration choices are made exclusively on the training datasets,
 286 as described in Section 4. The selected configuration is then fixed before
 287 evaluation on the held-out test datasets. The configuration selected by this
 288 procedure is reported in Section 5 together with its empirical evaluation.

289 3.7. Computational complexity

290 With dense pairwise computations, estimating the within-set neighborhood
 291 radii and cross-set nearest neighbors requires

$$O((n^2 + m^2 + nm)D)$$

292 operations.

293 The complete configuration space does not require the pairwise dissimilar-
 294 ities to be recomputed for every ALG configuration. Pairwise dissimilarity
 295 matrices are computed once and reused to evaluate the candidate local simi-
 296 larities and global similarities. The local-global weight is then applied without
 297 recomputing the pairwise dissimilarities.

298 This factorization makes configuration selection computationally feasible.

299 4. Experimental methodology

300 The experimental methodology separates configuration selection from final
 301 evaluation. All choices that define ALG* are made using the training datasets.
 302 The held-out test datasets are used only after the selected ALG configuration
 303 has been fixed.

304 4.1. Research questions

305 The experimental evaluation addresses five research questions:

- 306 • **RQ1 - Effectiveness:** How effectively do Initial ALG and training-
 307 selected ALG* distinguish similar from dissimilar embedding sets com-
 308 pared with the complete baseline suite, overall, across dataset families,
 309 and on neural-model activations?
- 310 • **RQ2 - Ablation:** For Initial ALG and ALG*, what is the contribu-
 311 tion of the local similarity, the global similarity, and their local-global
 312 combination?
- 313 • **RQ3 - Configuration Stability:** Is the configuration selected on
 314 the training datasets supported by stable training and held-out test
 315 rankings and by a broad near-optimal region?
- 316 • **RQ4 - Sensitivity:** How do ALG*, Initial ALG, and the baseline
 317 measures respond to controlled changes in local and global structure?
- 318 • **RQ5 - Efficiency:** How computationally efficient is ALG compared
 319 with the baseline measures, and what is the one-time cost of configura-
 320 tion selection?

321 4.2. Datasets

322 The benchmark resolver contains 250 candidate datasets from seven dataset
 323 families. Seven datasets are excluded before configuration selection: one image
 324 dataset has an incomplete computation and six time-series datasets contain a
 325 non-finite or unavailable ALG component. The benchmark therefore contains
 326 243 eligible datasets and 208,725 labeled embedding-set pairs, split at the
 327 dataset level into 60 training datasets (53,332 pairs) and 186 held-out test
 328 datasets (155,393 pairs). No dataset appears in both partitions.

329 Each dataset is converted into labeled pairs of embedding sets according
 330 to its native representation. Graph datasets use node and spectral descriptors;
 331 image datasets use local patches; OpenML and tabular datasets use class-
 332 conditional row or image-patch representations; text datasets use sentence or
 333 token representations; and time-series datasets use sliding-window represen-
 334 tations. The neural-model-activation dataset uses sets of hidden activations
 335 and belongs to the training partition. Each eligible dataset contributes up
 336 to 1,000 labeled embedding-set pairs in total, balanced between positive and
 337 negative pairs whenever possible, with smaller datasets contributing all valid
 338 pairs. The released dataset catalog records the source collection, source URL,

Table 3: Dataset-level training/test split.

Dataset family	Training	Test
Graphs	6	15
Images	5	12
OpenML	15	57
Tabular	2	1
Text	1	6
Time series	30	92
Neural-model activations	1	3
Total	60	186

339 construction rule, pair count, eligibility decision, partition, and redistribution
 340 status for every candidate dataset.

341 After configuration selection, the fixed ALG* configuration and the com-
 342 plete. For each source dataset, multilayer perceptrons with four hidden-layer
 343 architectures and five random seeds are fit on 5,000 samples from the of-
 344 ficial training split. Hidden activations are then extracted from a disjoint
 345 1,000-sample subset of the official test split. This produces 20 activation-set
 346 representations and 80 labeled pairs per dataset. These three datasets are
 347 not used for ALG configuration selection.

348 4.3. Threshold calibration

349 Each similarity measure produces a scalar value and requires a threshold
 350 for binary evaluation. No downstream classifier is trained for these scalar
 351 similarities; C2ST remains classifier-based by definition. Within each dataset,
 352 we use stratified five-fold cross-validation. For each fold, the threshold that
 353 maximizes the positive-label F1-score ($y = 1$) is selected on the other four
 354 folds and then applied without change to the held-out fold. The five held-out
 355 prediction vectors are pooled, and one positive-label F1-score is computed as

$$F1_{y=1} = \frac{2TP}{2TP + FP + FN}. \quad (20)$$

356 This is equivalent to micro-F1 restricted to label 1; it is not macro-averaged
 357 over labels 0 and 1. This protocol evaluates the discriminative ordering
 358 induced by the similarity measure while avoiding threshold selection on the
 359 evaluated fold.

360 Threshold calibration is distinct from configuration selection. For ALG,
 361 all 168,960 configurations are ranked by the unweighted mean of their positive-
 362 label F1-scores across the 60 training datasets. The highest-ranked configura-
 363 tion is then fixed before evaluation on the 186 held-out test datasets.

364 For every tunable baseline, candidate configurations are evaluated on
 365 the same 60 training datasets and selected by the same unweighted mean
 366 positive-label F1-score criterion. The selected configuration of each baseline is
 367 fixed before the held-out evaluation. Baselines without tunable configurations
 368 retain their fixed definitions. No configuration is selected separately for an
 369 individual dataset or dataset family, and no held-out test dataset is used
 370 to select a configuration. The local similarity, global discrepancy, geometry,
 371 normalization, scale multiplier, and local-global weight of ALG* remain fixed
 372 throughout the held-out evaluation.

373 4.4. Baselines and ALG variants

374 We distinguish four roles. ALG denotes the configurable formulation. Ini-
 375 tial ALG denotes its reference instance, introduced to motivate the local-global
 376 construction. ALG* denotes the single configuration selected on the training
 377 datasets and frozen before held-out evaluation. Internal ALG components
 378 and alternative ALG configurations are used only for ablation or configuration
 379 analysis; they are not additional proposed methods. All remaining comparison
 380 methods are baselines.

381 The baseline suite contains 38 literature baselines and two standard
 382 pooled baselines. The literature baselines cover neighborhood-based measures,
 383 precision-recall and Density/Coverage measures, point-set distances, distri-
 384 bution distances, kernel two-sample measures, optimal-transport measures,
 385 classifier two-sample tests, and geometric or topological measures. The two
 386 standard pooled baselines are included explicitly because they provide the
 387 simplest global comparison after mean pooling:

$$S_{\cos}^{\text{pool}}(A, B) = s_{\cos}(\mu_A, \mu_B), \quad (21)$$

$$S_{\text{Euc}}^{\text{pool}}(A, B) = -\|\mu_A - \mu_B\|_2. \quad (22)$$

388 The negative sign in the Euclidean baseline only converts a distance into a
 389 similarity so that larger values consistently indicate greater similarity. These
 390 two baselines test whether comparing the mean embeddings alone is sufficient.

391 Table 4 separates methodological family from original application domain.
 392 The distinction matters because the benchmark deliberately places methods

393 developed for generative-image evaluation, point-cloud reconstruction, image
 394 matching, spectral clustering, and general statistical two-sample comparison
 395 in the same heterogeneous evaluation. “Domain-oriented” describes the setting
 396 in which a method was introduced; it does not restrict the datasets on which
 397 the method is evaluated here.

398 Pooled cosine has two roles in the analysis that refer to the same computed
 399 similarity: it is a standard pooled baseline in the baseline comparison and it
 400 is exactly the global-only component of Initial ALG in the matched ablation.
 401 It is therefore not recomputed or treated as a different method in the ablation.
 402 The local-only component of Initial ALG is the symmetric cosine overlap
 403 with $k = 1$ defined in Eq. (7); it remains an internal ALG component rather
 404 than a baseline. Pooled Euclidean is not a component of Initial ALG and is
 405 included only as a standard pooled baseline.

406 The fixed method taxonomy therefore contains the Initial ALG reference
 407 instance, ALG^* , 40 baseline outputs, and six internal ALG components or
 408 alternative configurations. In the completed campaign, 40 baselines, Initial
 409 ALG, and ALG^* are available on all 186 held-out test datasets. Gaussian
 410 Fréchet is available on 133 test datasets and is never imputed. We report its
 411 coverage explicitly and retain a separate 133-dataset common-subset analysis
 412 in the results. The supplied 136-dataset graphic additionally includes the three
 413 neural-model-activation datasets and is not used to represent the 133-dataset
 414 confirmatory subset. No baseline is removed on the basis of its observed
 415 F1-score.

416 4.5. *Statistical analysis*

417 The unit of statistical analysis is the dataset. Unless explicitly stated
 418 otherwise, reported means are unweighted means of the per-dataset positive-
 419 label F1-scores; they are not obtained by pooling predictions across datasets.
 420 We report mean and median positive-label F1-score, fractional dataset wins,
 421 strict and tied dataset wins, and paired wins/ties/losses. For each ALG-versus-
 422 baseline comparison, the mean paired F1-score difference is accompanied by
 423 a 95% confidence interval estimated from 10,000 paired bootstrap resamples.
 424 We also use the Wilcoxon signed-rank test and apply Holm correction over
 425 the complete set of 40 baseline comparisons separately for Initial ALG and
 426 ALG^* . Pairing is performed at the dataset level so that differences in dataset
 427 difficulty do not confound the comparison.

428 The ranking of all configurations on the held-out test datasets is used only
 429 as a post-hoc stability analysis. It is not used to redefine ALG^* or to make a

Table 4: Organization of the 40 baselines by methodological family and original application domain. Every baseline remains in the overall and cross-family comparisons.

Methodological family	Included baselines	Original application domain
Distribution and kernel comparison	Energy distance, MMD-RBF, KID, Gaussian Fréchet, C2ST	Statistical two-sample testing or generative-model evaluation
Optimal transport	Sliced Wasserstein, Earth Mover/Wasserstein, Sinkhorn divergence, Gromov-Wasserstein	General distribution comparison and geometric matching
Adaptive neighborhoods	Local-scaling affinity	Spectral clustering
Fidelity and diversity	PRD F1-max, manifold precision/recall, PRDC variants, Coverage, clipped Density/Coverage, Precision-Recall Cover, alpha precision, beta recall, and authenticity	Generative-model evaluation
Point-set geometry	Chamfer, modified Hausdorff, Hausdorff-95, Hausdorff, Density-Aware Chamfer, and point-cloud F-score	Image matching or 3D point-cloud reconstruction
Mutual-neighbor and topology	Best Buddies, persistence bottleneck, and Geometry Score	Image matching or generative-model geometry
Pooled representation	Pooled cosine and Pooled Euclidean	General-purpose embedding comparison

second configuration selection.

4.6. Controlled perturbation experiments

The held-out benchmark measures discrimination on real datasets but does not identify which distributional changes each similarity measure detects. We therefore use controlled perturbation experiments following the diagnostic logic of Naeem et al. [4]. The full campaign contains 145 perturbation conditions and 128 random seeds, corresponding to 18,560 condition-seed evaluations. The conditions include translation, variance change, outliers, cardinality imbalance, dimension, duplicates, nested and non-convex supports, rings, collapsed arcs, filled disks, and sequential or simultaneous mode removal. A dedicated protocol also evaluates Gaussian mixtures, mean shifts, outliers, and mode removal.

The controlled perturbation experiments are not used for ALG configuration selection. They are diagnostic experiments: as perturbation severity increases, a similarity measure is expected to change monotonically when the perturbation corresponds to information represented by that measure. Failure to respond is therefore interpreted as an invariance or limitation of the underlying geometry, not as a configuration-selection result.

5. Results

We evaluate 168,960 ALG configurations on 243 benchmark datasets: 60 training datasets and 186 held-out test datasets. The frozen configuration and the baseline suite are additionally evaluated on three neural-model-activation datasets.

5.1. Selected ALG configuration

Configuration selection is performed only on the 60 training datasets. Among the configurations in the completed search campaign, the selected configuration combines local scaling with Manhattan distance and $k = 3$ with a normalized cosine-centroid global similarity. Its local-global weight is $\lambda = 0.7$:

$$\text{ALG}^*(A, B) = 0.7 L_{\text{LS}, L_1, k=3}(A, B) + 0.3 \exp \left[-\frac{\delta_{\cos}(\mu_A, \mu_B)}{2} \right] \quad (23)$$

Thus, the selected configuration gives greater weight to the adaptive local component while retaining a non-zero global contribution. This configuration is fixed before evaluation on the 186 held-out test datasets.

462 5.2. RQ1: Effectiveness

463 Table 5 reports the principal comparison on the 186 held-out test datasets.
 464 Figure 2 contains ALG*, Initial ALG, and the 40 baselines with complete
 465 test coverage, including Pooled cosine and Pooled Euclidean. Internal ALG
 466 similarities are not included in this comparison. The means in the figure
 467 and Table 5 use different dataset populations and should not be compared as
 468 identical estimates.

469 ALG* obtains the highest mean positive-label F1-score across datasets,
 470 0.7556. Initial ALG reaches 0.7507, and Energy distance, the strongest
 471 literature baseline with complete test coverage, reaches 0.7498. The difference
 472 between ALG* and Initial ALG is an internal comparison and measures
 473 the effect of configuration selection within ALG. The baseline comparison
 474 evaluates ALG against the 38 literature baselines and the two standard pooled
 475 baselines. Pooled cosine and Pooled Euclidean reach mean positive-label F1-
 476 scores of 0.7392 and 0.7386, respectively.

Table 5: Summary of the held-out comparison. The complete baseline ranking is reported in Figure 2.

Method	n	Mean $F1_{y=1}$	Median $F1_{y=1}$
ALG*	186	0.7556	0.6871
Initial ALG	186	<u>0.7507</u>	0.6813
Energy distance	186	0.7498	<u>0.6852</u>
Pooled cosine	186	0.7392	0.6698
Pooled Euclidean	186	0.7386	0.6707

477 The paired comparison gives the same ordering. Against Energy distance,
 478 ALG* has a mean paired F1-score difference of 0.0058, with a 95% bootstrap
 479 confidence interval of [0.0013, 0.0104]. ALG* is higher on 116 datasets and
 480 lower on 67. The Holm-corrected Wilcoxon p -value is 0.0013. Initial ALG
 481 differs from Energy distance by 0.0010 on average, with a bootstrap interval
 482 that includes zero. This distinction is important: the stronger statistical
 483 result is associated with the configuration selected on the training datasets,
 484 not with the initial formulation alone.

485 The same fixed ALG* configuration is then examined by dataset family;
 486 no family-specific configuration is selected. Figure 4 retains every comparison
 487 method so that variation between methodological families is visible rather

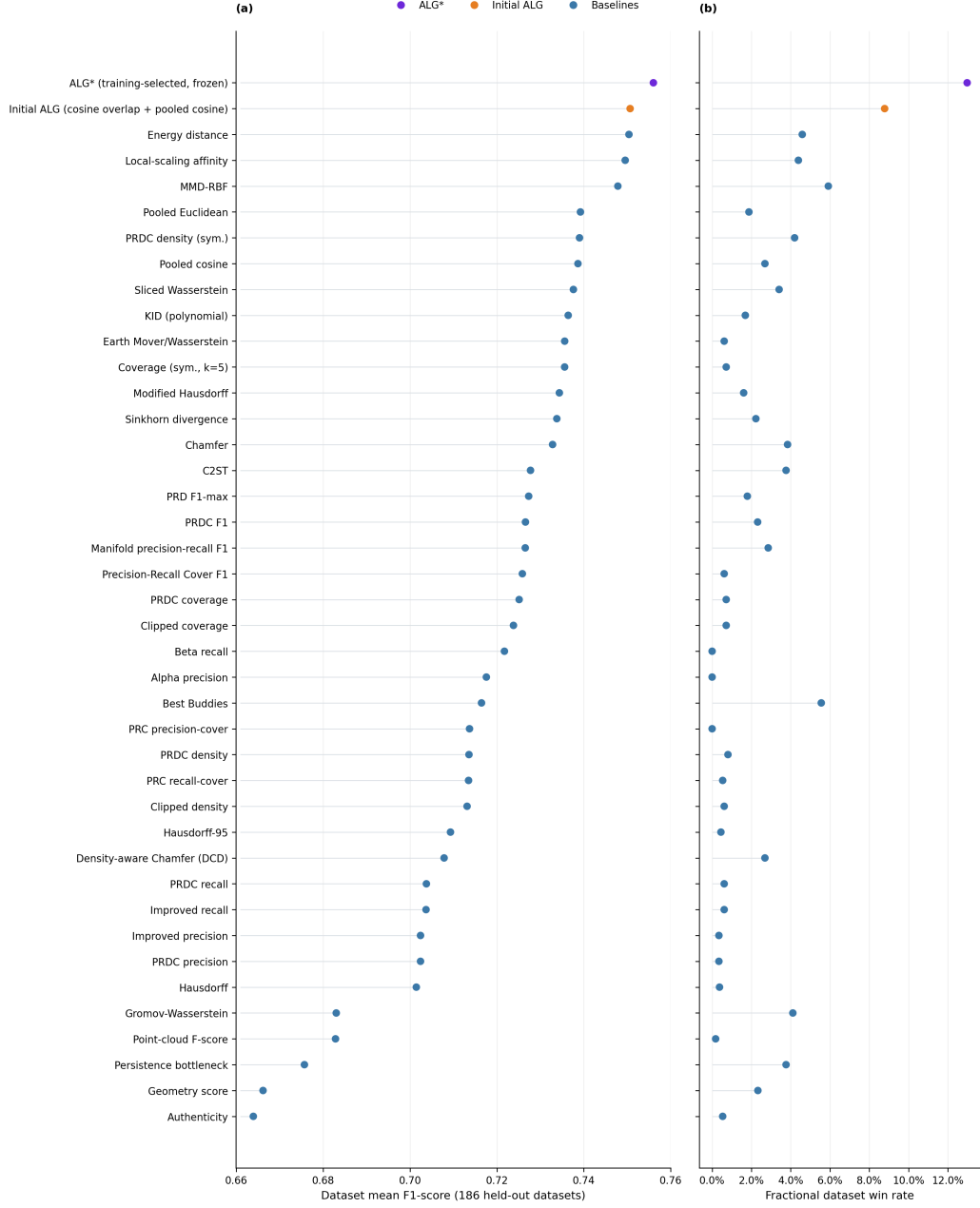


Figure 2: Mean positive-label F1-score and fractional dataset win rate across 186 evaluation datasets. ALG* remains fixed from the 60 training datasets. This expanded overview is distinct from the 186-dataset confirmatory results in Table 5.



Figure 3: Paired F1-score differences and dataset win rates against all 40 baseline measures. Gaussian Fréchet is evaluated on the 133 test datasets for which it is available; all other comparisons use 186 test datasets.

Table 6: Paired comparison with Energy distance, the strongest external baseline with complete test coverage. Holm correction is computed over all 40 baseline comparisons separately for each ALG method. The best value is bold; the second-best value is underlined.

ALG method	$\Delta F1$	95% CI	W/T/L	Holm p
ALG*	0.0058	[0.0013, 0.0104]	116/0/67	0.0013
Initial ALG	<u>0.0010</u>	[-0.0047, 0.0065]	96/7/80	<u>0.2431</u>

488 than restricted to a hand-picked subset. Table 7 reports the two ALG methods
 489 for reference.

490 ALG* reaches mean positive-label F1-score 0.6661 on graphs, 0.6684 on
 491 images, 0.8738 on OpenML, 0.9980 on the single tabular dataset, 0.7228 on
 492 text, and 0.7078 on time series. The variation across families is substantial,
 493 and the selected ALG configuration is not the highest-scoring method in
 494 every family. The relevant observation is that the same selected configuration
 495 remains close to the leading values across dataset families without test-time
 496 re-selection.

Table 7: The two ALG methods across the held-out dataset families. The complete method-by-family comparison is shown in Figure 4. The best value in each row is bold; the second-best value is underlined.

Dataset family	n	ALG*	Initial ALG
Graphs	15	0.6661	<u>0.6653</u>
Images	12	0.6684	<u>0.6663</u>
OpenML	57	0.8738	<u>0.8676</u>
Tabular	1	<u>0.9980</u>	0.9990
Text	6	<u>0.7228</u>	0.7357
Time series	92	0.7078	<u>0.7016</u>

497 The three neural-model-activation datasets provide a separate transfer
 498 check for the frozen configuration. ALG* reaches mean positive-label F1-
 499 score 0.785 on these datasets, compared with 0.746 for Initial ALG (values
 500 rounded from the additional evaluation). Other baselines also achieve higher
 501 values on these representations; the complete comparison is included in the
 502 neural-model column of Figure 4.

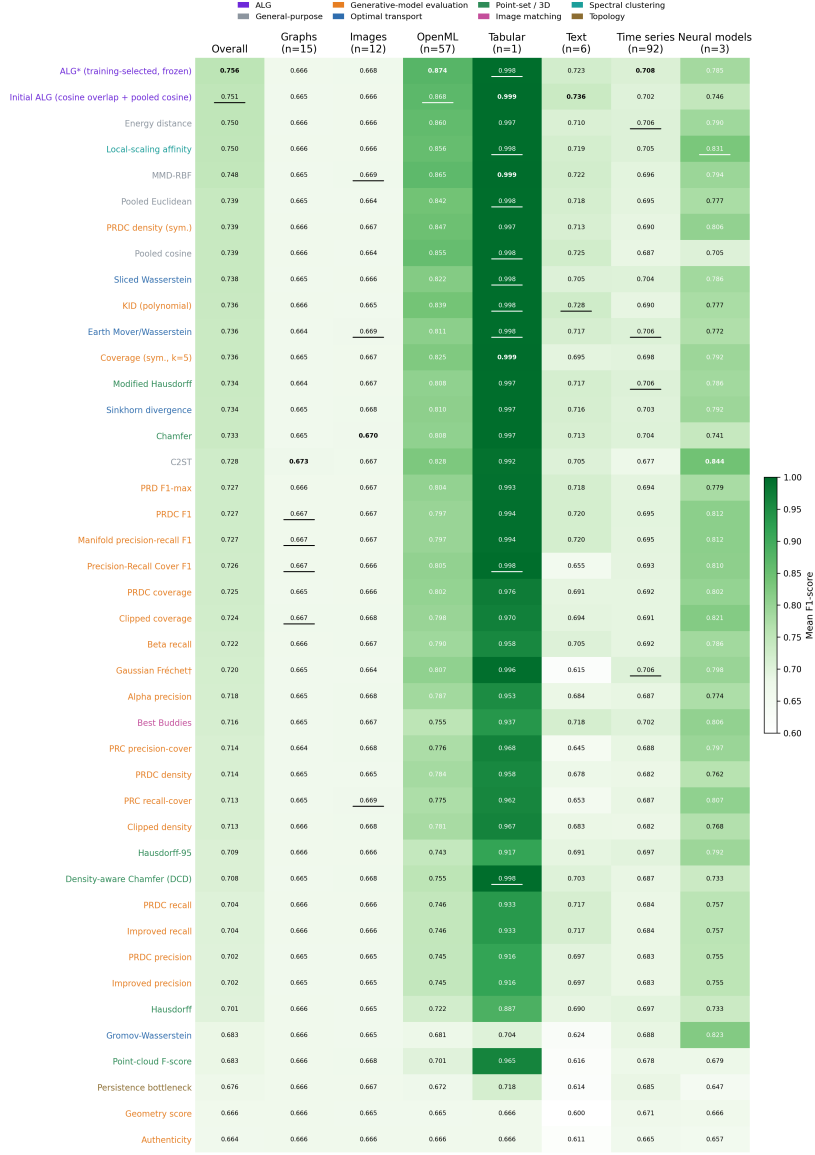


Figure 4: Mean positive-label F1-score by dataset family for all comparison methods. Graphs, images, OpenML, tabular data, text, and time series cover 186 held-out benchmark datasets; the neural-model column reports the three additional datasets. The Overall column of this figure includes all 186 datasets and therefore differs from the 186-dataset estimates in Table 5. The heatmap encodes F1-score, and colored method labels indicate the original application domain; gray denotes general-purpose methods. Gaussian Fréchet is available on 133 of the 186 held-out benchmark datasets; its displayed means use available datasets only.

503 *Discussion..* The held-out comparison supports two distinct conclusions.
504 First, Initial ALG reaches a mean positive-label F1-score of 0.7507, compared
505 with 0.7498 for Energy distance, 0.7392 for Pooled cosine, and 0.7386 for
506 Pooled Euclidean. However, its paired difference from Energy distance is
507 small and its confidence interval includes zero. Second, configuration selection
508 on the training datasets increases the mean positive-label F1-score to 0.7556.
509 The paired difference between ALG* and Energy distance is positive under
510 both the bootstrap analysis and the Holm-corrected Wilcoxon test.

511 The magnitude of the improvement remains small and should not be
512 interpreted as universal dominance over every baseline or dataset. Different
513 methods lead on different dataset families, and the neural-model-activation
514 evaluation also shows that ALG* is not uniformly best. The evidence supports
515 the more specific conclusion that one training-selected local-global configura-
516 tion transfers across heterogeneous held-out datasets and improves the mean
517 positive-label F1-score across datasets over the strongest complete-coverage
518 external baseline.

519 5.3. RQ2: Ablation

520 Table 8 compares each ALG method with its corresponding local-only and
521 global-only similarity on the same 186 held-out test datasets. For Initial ALG,
522 the local-only term is symmetric cosine overlap with $k = 1$ and the global-only
523 term is exactly the Pooled cosine baseline of Eq. (22). The global-only Initial
524 ALG value therefore appears both in the baseline comparison and in this
525 ablation by design; it is the same similarity measure and the same result.
526 For ALG*, the local-only similarity is local scaling with Manhattan distance
527 and $k = 3$, while the global-only similarity is the normalized cosine-centroid
528 component of Eq. (23).

529 The ablation shows that the local similarity obtains the higher mean
530 positive-label F1-score of the two individual similarities for each ALG method.
531 However, the local-global combination improves the mean positive-label F1-
532 score across datasets over the local similarity alone. This result is consistent
533 with the configuration frequencies in Figure 6, where intermediate values of λ
534 dominate the high-performing region.

535 *Discussion..* The matched ablation identifies the local similarity as the
536 stronger individual component of both ALG instances. Nevertheless, the
537 local-global combination obtains the highest mean positive-label F1-score in
538 each block. The global similarity retains discriminative information on its

Table 8: Matched local-only and global-only ablation. Wins/ties/losses compare each component with the corresponding local-global ALG similarity. Within each ALG block, the best score is bold and the second-best score is underlined.

ALG method	Variant	Mean $F1_{y=1}$	Median $F1_{y=1}$	W/T/L
Initial ALG	Sym. cosine overlap ($k = 1$)	<u>0.7453</u>	0.6887	81/9/93
Initial ALG	Pooled cosine	0.7392	0.6698	53/19/111
Initial ALG	Local-global	0.7507	<u>0.6813</u>	-
ALG*	Local scaling (L_1 , $k = 3$)	<u>0.7504</u>	0.6874	63/19/101
ALG*	Normalized cosine centroid	0.7391	0.6689	56/11/116
ALG*	Local-global	0.7556	<u>0.6871</u>	-

own and contributes complementary information when combined with the local similarity. The gain is modest, so the ablation supports complementarity rather than claiming that fusion is superior on every individual dataset.

5.4. RQ3: Configuration Stability

The configuration space contains 168,960 alternatives, so the stability of configuration selection must be evaluated explicitly. When all configurations are ranked retrospectively on the 186 held-out test datasets, the configuration selected on training ranks 196th, corresponding to the top 0.116% of the configuration space. The retrospective test optimum reaches mean positive-label F1-score 0.7582, only 0.0026 above the selected score of 0.7556. The Spearman correlation between training and test mean-positive-F1 rankings is $\rho = 0.900$.

The training optimum is also part of a broad near-optimal region. The difference between the first and second training configurations is 1.535×10^{-5} . Three configurations are within 0.0001 of the training maximum, 55 are within 0.001, 2,422 are within 0.005, and 25,701 are within 0.01. Figure 5 shows the training-test association in the expanded 186-dataset evaluation and the size of the training near-optimal region. The result supports the stability of a family of high-performing local-global configurations rather than the uniqueness of one hyperparameter tuple.

The high-performing region also provides information about the selected design. Among the top 1,000 training configurations, local scaling is used in

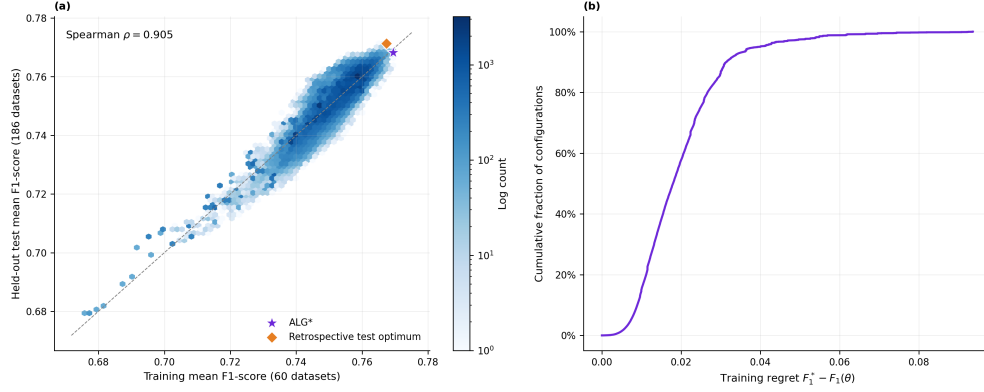


Figure 5: Configuration-selection diagnostics over 168,960 ALG configurations. Panel (a) plots the 60-training-dataset means against the expanded 186-dataset evaluation means; Spearman $\rho = 0.905$). Panel (b) shows the cumulative proportion of configurations within a given training regret. The 186-dataset analysis reported in the text is a separate evaluation population.

561 91.9% of the local similarities. Among configurations whose local similarity
562 has a neighborhood parameter, $k = 3$ is used in 76.9%. Manhattan, cosine,
563 and Euclidean geometries all remain represented. Centroid and Energy dis-
564 crepancies account for 48.1% and 41.1% of the global similarities, respectively,
565 while MMD accounts for 10.8%. The local-global weights are concentrated
566 between 0.3 and 0.8, so the high-performing region does not collapse to a
567 local-only or global-only similarity.

568 *Discussion..* A search over 168,960 configurations creates a risk of selecting an
569 isolated training optimum. The held-out analysis does not show this pattern:
570 the selected configuration ranks 196th retrospectively, the test optimum is
571 only 0.0026 higher in mean positive-label F1-score, the training/test rank
572 correlation is $\rho = 0.900$, and thousands of configurations lie close to the
573 training optimum.

574 These results do not establish that Manhattan distance, $k = 3$, expo-
575 nential normalization, or $\lambda = 0.7$ are universally optimal. They show that
576 configuration selection identifies a broad high-performing region in which
577 an adaptive local similarity is generally combined with a non-zero global
578 similarity.

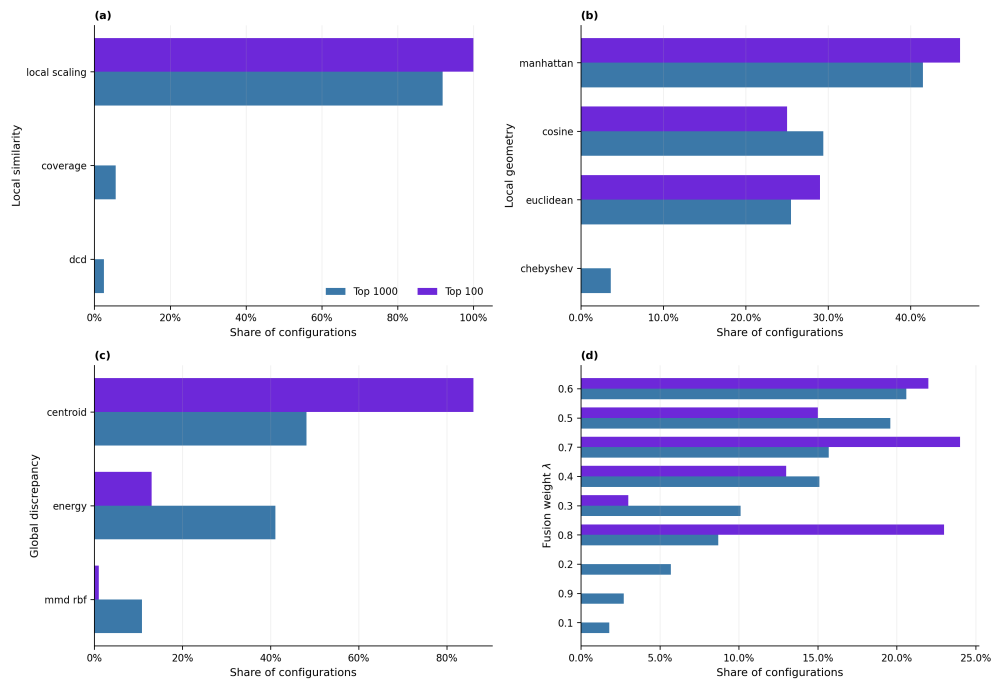


Figure 6: Configuration frequencies among the top 100 and top 1,000 training configurations. Intermediate local-global weights and local scaling are common throughout the high-performing region.

5.5. RQ4: Sensitivity

The controlled perturbation experiments characterize the response of the similarity measures when the distributional difference is known. Figure 7 reports Gaussian translation, nested-support scaling, and non-convex ring perturbations. It compares the frozen ALG* configuration with representative baselines. Each curve is min-max normalized within method and panel; consequently, the figure compares response shape rather than raw similarity values.

All three horizontal axes are oriented so that perturbation severity increases from left to right. ALG* decreases monotonically across the three displayed experiments. These experiments do not establish that one response shape is universally preferable; they show how the selected local and global geometries determine the changes to which ALG* is sensitive.

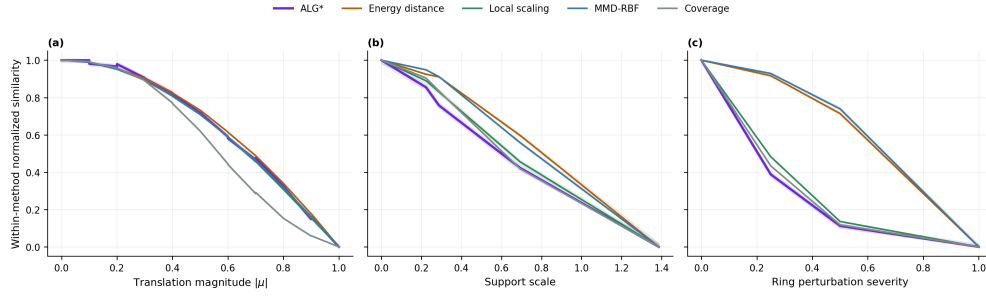


Figure 7: Controlled perturbation response of ALG* and representative baseline measures. The horizontal axes are oriented so that perturbation severity increases from left to right. Similarities are min-max normalized within each method and panel.

Figure 8 complements the response curves with an exhaustive comparison of ALG* and all 40 baselines. For each method and perturbation, the cell reports the negative Spearman correlation between perturbation severity and mean similarity. A value near one denotes a monotone decrease in similarity, a value near zero denotes a weak monotone response, and a negative value denotes the opposite orientation. For nested support, severity is the absolute log-ratio from equal support. This matrix summarizes sensitivity rather than discriminative F1-score and does not identify one response profile as universally preferable.

Discussion.. The controlled experiments show that ALG* responds monotonically to the displayed perturbations. Its response is determined jointly by its

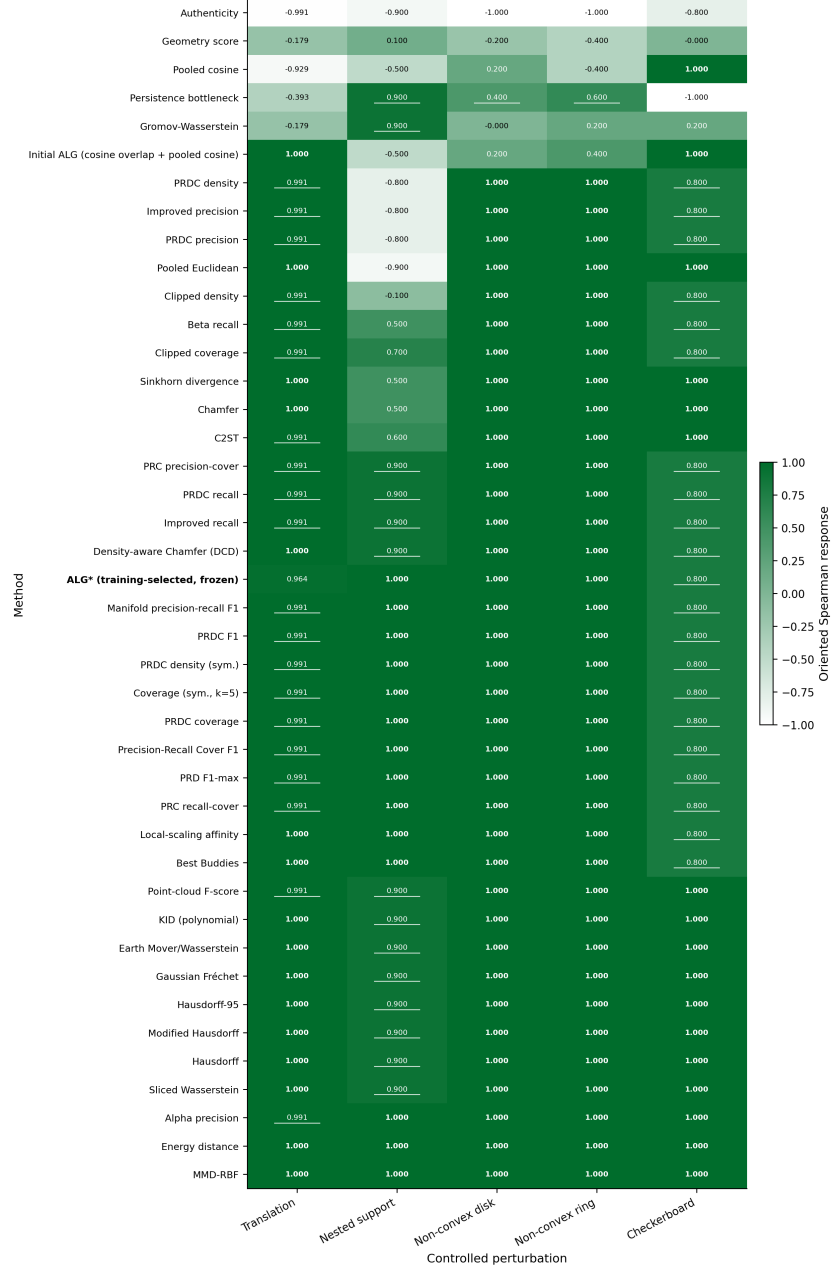


Figure 8: Controlled sensitivity of ALG* and all 40 baselines across five multi-level perturbations. Cells show the oriented Spearman response computed from the mean similarity at each perturbation level.

603 adaptive local-scaling component and its normalized global cosine-centroid
 604 component. The comparison also confirms that different baseline families
 605 respond to different perturbation geometries. These results characterize the
 606 sensitivity profile of the frozen ALG* configuration rather than establishing
 607 universal superiority under every controlled perturbation.

608 5.6. RQ5: Efficiency

609 To answer RQ5, we measure both deployment-time efficiency and the
 610 one-time cost of configuration selection. Deployment-time measurements
 611 are performed on the same 155,393 embedding-set pairs from the 183 test
 612 datasets. For ALG* and Initial ALG, each similarity is recomputed end-to-end
 613 from the raw embedding sets without reusing cached distances or framework
 614 components. Each method is executed independently three times per pair,
 615 and the median elapsed time is retained before aggregation.

616 ALG* requires an average of 0.780 ms per embedding-set pair, compared
 617 with 1.046 ms for Initial ALG. The selected configuration therefore reduces
 618 the end-to-end runtime by 25.4%, corresponding to a $1.34\times$ speedup over
 619 the initial formulation. Figure 9 reports these measurements together with
 620 the recorded runtimes of all 40 baselines and the explicitly labeled diagnostic
 621 computations. The diagnostic totals and ALG-family totals are internal
 622 measurements and are not additional baselines.

623 Among the 40 baselines, ALG* is faster than 30 and slower than 10. The
 624 methods that remain faster are predominantly simple pooled or geometric
 625 comparisons, whereas several distributional, optimal-transport, topological,
 626 and classifier-based metrics require substantially more computation. Thus,
 627 ALG* improves both effectiveness and runtime relative to Initial ALG, but it
 628 is not the fastest method in absolute terms.

629 Across the 243 eligible datasets, cumulative component computation
 630 requires 7.39 task-hours and evaluation of the 168,960 scalar local-global
 631 combinations requires 3.27 task-hours. These values are sums over parallel
 632 tasks rather than Slurm wall-clock time. The median component-computation
 633 time is 8.68 s per dataset and the median configuration-evaluation time is
 634 54.27 s per dataset.

635 *Discussion..* The efficiency results distinguish the one-time cost of selecting
 636 ALG* from its deployment-time cost. The exhaustive search is made feasible
 637 by computing reusable local and global components during training. Once
 638 the configuration is frozen, however, ALG* is evaluated directly and does not

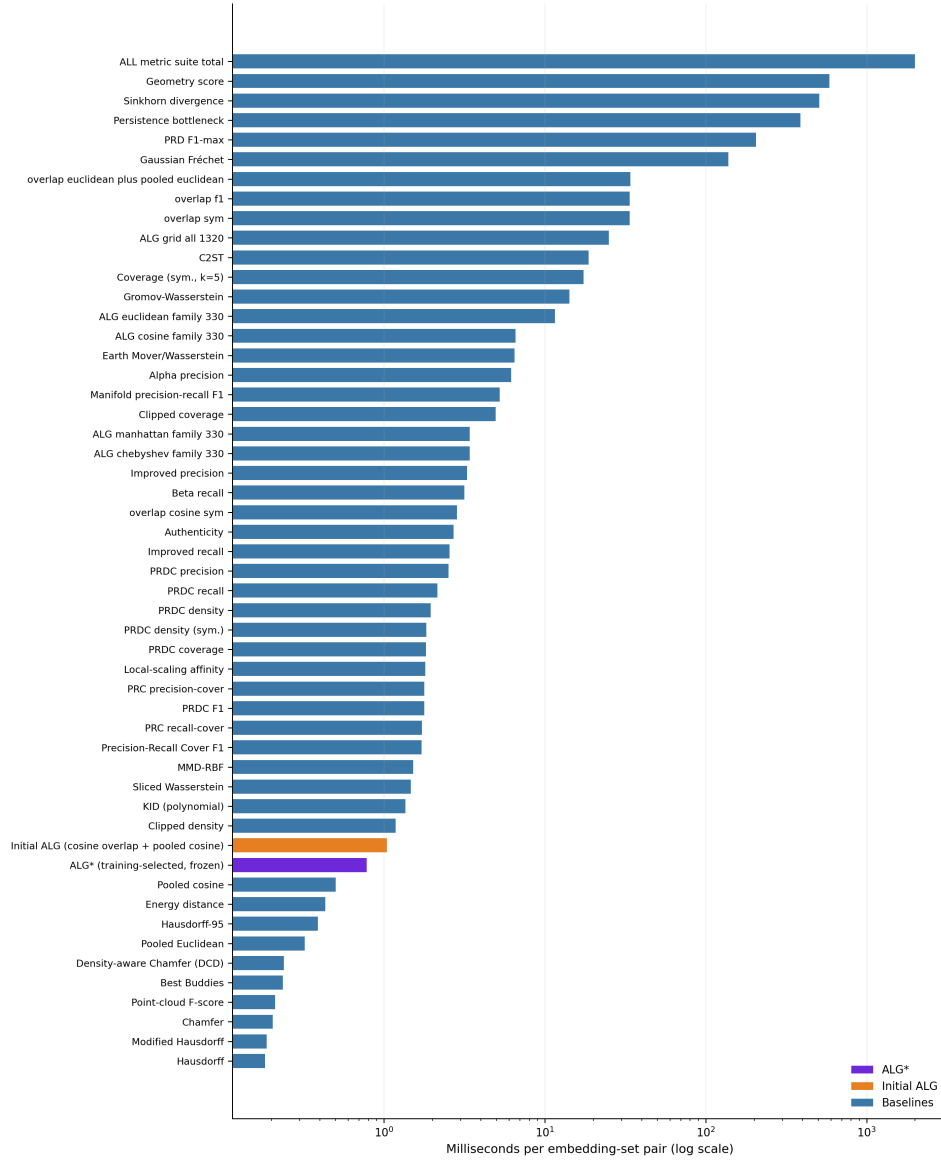


Figure 9: Measured end-to-end runtime per embedding-set pair on 155,393 pairs from the 183 test datasets. ALG* and Initial ALG are recomputed from the raw embedding sets without cached components; their reported values aggregate the median of three independent executions per pair. The figure also includes all 40 baselines and explicitly labeled diagnostic computations. Diagnostic totals and ALG-family totals are not additional baselines. The horizontal axis uses a logarithmic scale.

require the complete component grid. Its measured runtime of 0.780 ms per pair shows that the selected local–global formulation remains inexpensive at deployment and is 25.4% faster than Initial ALG. Nevertheless, simpler pooled and geometric baselines can be faster, so the result establishes a favorable effectiveness–efficiency trade-off rather than a universal runtime advantage.

5.7. Limitations

The test datasets are not balanced across dataset families. The held-out split contains 92 time-series datasets and 57 OpenML datasets, but only six text datasets and one tabular dataset. Dataset averaging prevents datasets with many embedding-set pairs from dominating the result, but it does not give equal weight to each dataset family. In addition, datasets from the same archive, such as UCR or OpenML, can share construction characteristics; the 186 held-out datasets should therefore not be interpreted as 186 independent application domains.

Gaussian Fréchet is available on only 133 held-out test datasets. We report its coverage explicitly and retain the strict 133-dataset common-subset comparison rather than imputing missing values. The retrospective test optimum over the 168,960 ALG configurations is used only to analyze selection stability and must not be interpreted as a confirmatory competitor. The runtime comparison also combines component-based ALG timing with monolithic timing for most baseline measures, so it is not a controlled systems benchmark.

Finally, the binary threshold is calibrated within each dataset. This protocol isolates the ordering induced by the similarity measure, but it does not test whether one universal decision threshold transfers across datasets. Applications that require a fixed threshold need a separate calibration experiment.

6. Conclusion

We define Adaptive Local-Global similarity (ALG) for the comparison of finite unordered embedding sets. ALG combines an adaptive local similarity, whose scale is estimated from within-set nearest neighbors, with a normalized global similarity. The local similarity, global discrepancy, geometry, normalization, and local-global weight remain explicit configuration choices.

The selected ALG configuration is chosen using 60 training datasets and is then evaluated without further configuration selection on 186 held-out test datasets. It reaches a mean positive-label F1-score across datasets of 0.7556, compared with 0.7498 for Energy distance, the strongest literature baseline

674 with complete test coverage. The matched ablation shows that the local and
675 global similarities provide complementary information, and the controlled
676 perturbation experiments show that the response of ALG depends on the
677 geometry encoded by its local and global similarities.

678 The main result is therefore not that one geometry is universally optimal.
679 Rather, a local-global similarity can be selected without test leakage and can
680 retain competitive performance across heterogeneous dataset families while
681 keeping its geometric assumptions explicit. This makes ALG applicable as a
682 similarity layer in information-system and machine-learning pipelines that
683 compare sets of learned representations.

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